

Individual Assignment 5

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2026-05-10

1. Set Up

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/zeldareif/Desktop/github/week-05_spring-2026_aquatic-inverts

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(snakecase)

# wrap axis labels
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic info
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

Rows: 50 Columns: 15

-- Column specification -----

Delimiter: ","

chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)
```

2. Cleaning

sites object

Goal: create a new object from `sites` thst only includes NCOS sites

```
# creating new object
ncos_sites <- sites |>
  # filtering to only include ncos quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

clean taxon_list

Goal: convert character strongs in existing `taxon_list` data frame into snakecase

```
taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean water_quality

Goal: create a data frame with mean pH, mean DO (mg/L) and mean salinity (ppt) for each data and site

- create a new object from water_quality
- clean column names use janitor
- only include surveys from quarterly ncos sites
- create a new column called date_site to join with later - - data frames
- group by date_site
- calc median pH, salinity and dissolved oxygen
- ungroup
- filter out outliers

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

`summarise()` has regrouped the output.

i Summaries were computed grouped by date, site, and date_site.

i Output is grouped by date and site.

i Use `summarise(.groups = "drop\_last")` to silence this message.

i Use `summarise(.by = c(date, site, date\_site))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

clean aq_ins

Goal: create long format data frame in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality info for each survey.

- create new object
- clean column names
- filter to only include a quarterly NCOS sites
- select columns
- filter to only include “filtered beaker” sample
- convert data frame to long format
- group by site, date and taxon
- calculate avg. abundance per liter (sum counts divided by 7.5 liter)
- create new column called date_site to join with water
- quality
- join data frame
- join with taxon list

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
```

```

      values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
        date_site,
        med_ph,
        med_sal,
        med_do),
  by = "date_site"
) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
      site_full = fct_rev(site_full))

```

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Problems

1.